CDSD bloc 2
Cyril Braguy

Speed dating with TINDER

Exploratory Data Analysis

Github project : cyrilbraguy/Bl2_Tinder

Pitch and GOAL

Pitch

- Tinder, is facing declining matches and wants to understand **what makes people interested in each other.**
- A speed dating experiment has then been launched
- Four-minute dates in 21 waves with questionnaire data teach us about
  - people's life informations, habits , lifestyle
  - ratings of partner on six attributes
  - decision on second date

Goal

- Use the dataset, through descriptive statistics and visualizations to identify **what drives people to accept a second date.**
- Explore questions such as attractiveness versus its real impact, shared interests versus shared background, self-perception accuracy, and date order.
- Deliver a notebook with statistics, charts, and interpretations.

Plan of analysis

- Load and clean data
- EDA of speed daters & waves
- Analysis of questionnaires
- Conclusion : **what makes people interested in each other for a second date**

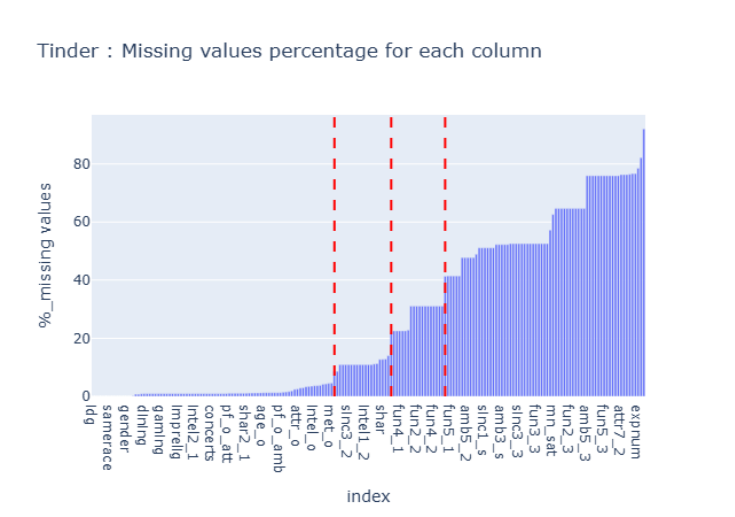
1 Load data

8378 lines 195 columns, 21 waves, 551 people

missing values : 26.4%

96 columns usable :

- 13 → 17 columns full
- 72 → 69 columns with less 5% missing values: imputation possible AVG/median or mode
- Not possible above : imputation too risky for data quality



Clean data:

- Data types : int, bool, float, categorial
- Columns rename for better intelligibility
- Replace columns dictionaries for categorical variables
- Ratings update on same scale : 1-10 -> sum_100 ; handle missing descriptor
- Aggregate data of persons in a DataFrame : perso_df
- Feature engineering : additional indicators :
  - success_rate,
  - decision_rate,
  - decision_partner_rate,
  - match_rate...
- compute stats per wave : synth1

2 EDA of speed daters

Population is mainly between 20 & 30 years, max age : women : 19-55, men : 18-42

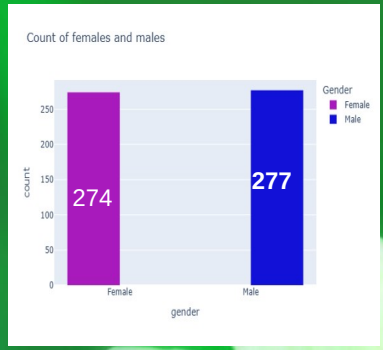
Age repartition : women younger than men,

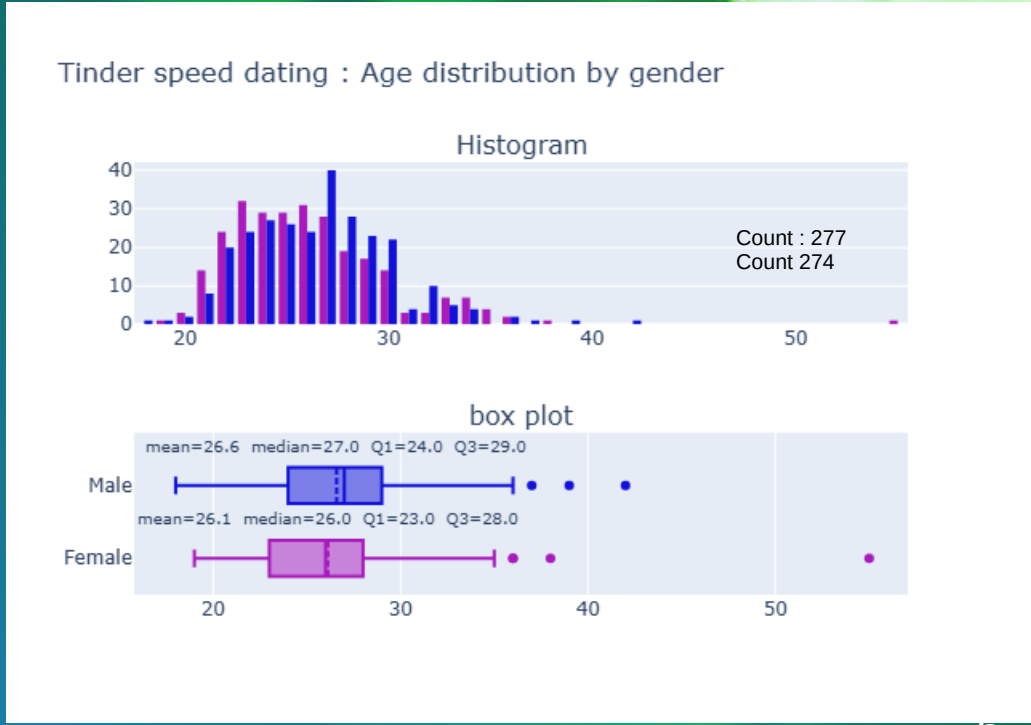
1 year younger in average

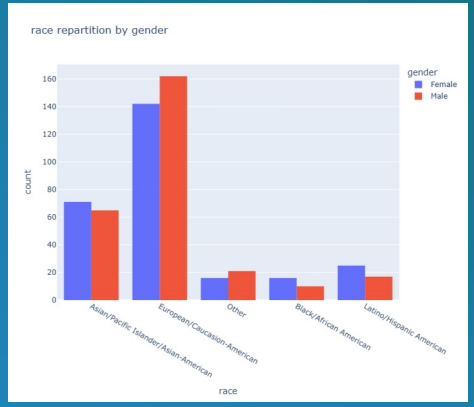
Each people is present in one single wave

Race European is over represented

Asian is second





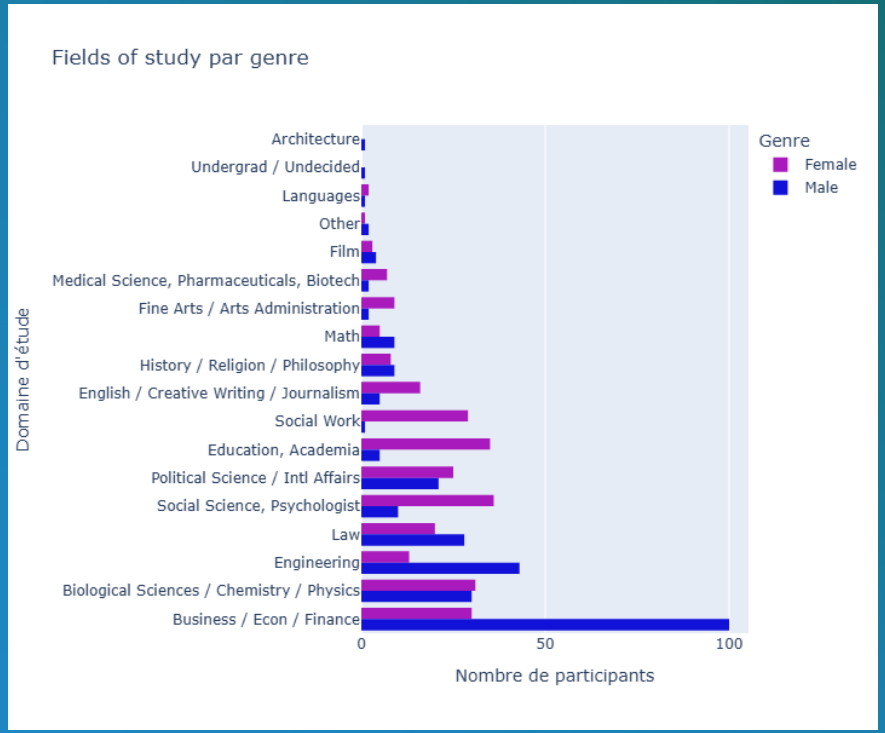


2 EDA of speed daters

Field of study highly gendered:

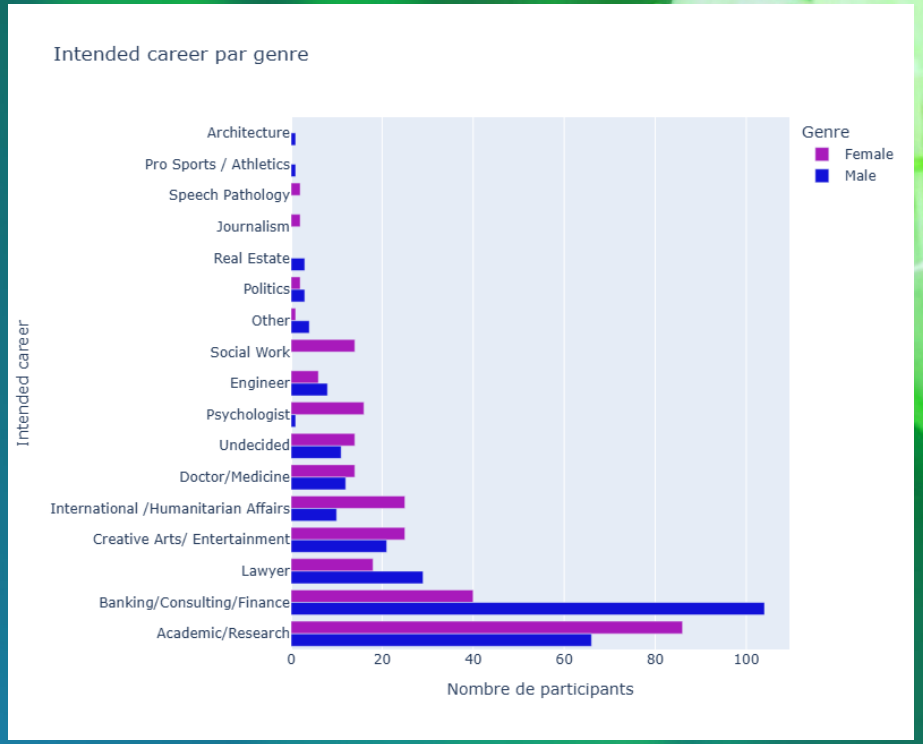
Men : **mode** : business/ Finance, Engineering, Bio sciences, law

Women : Social sciences, Education, social work



Finaly Intended career :

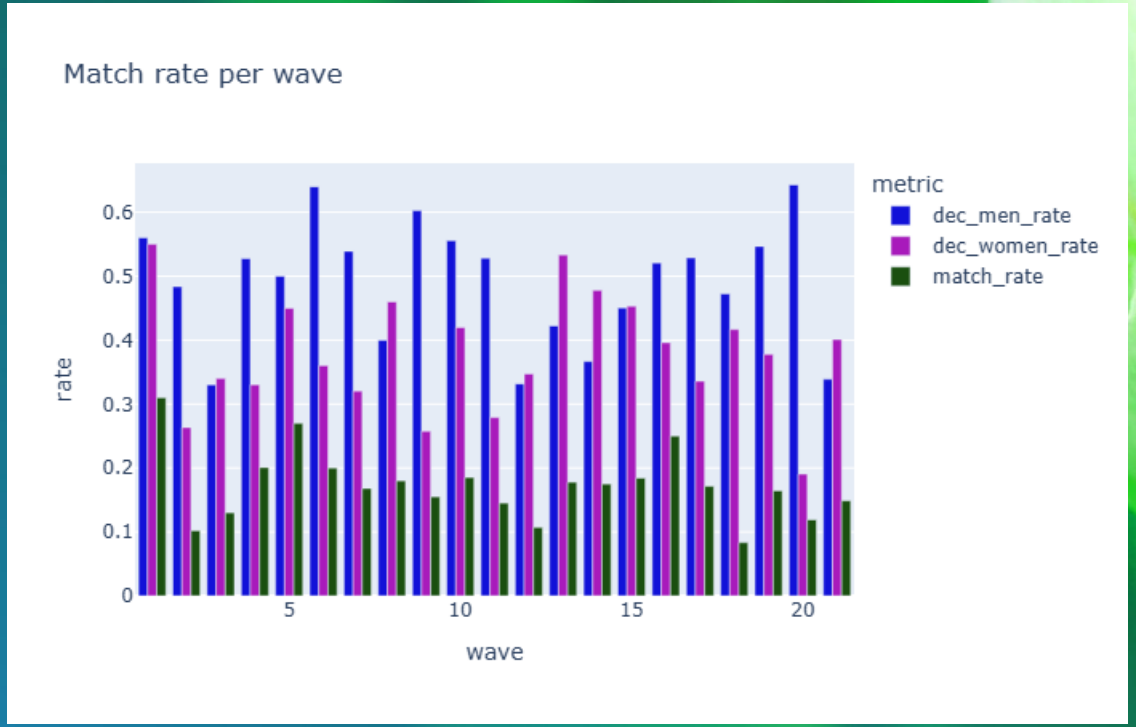
Miss of coherence with field of study
Bias : 2 marked modes:
- Men : banking / finance ,academic research
- Women, Academic research, banking / finance and then huanitarian affairs or creative arts ...



2 EDA of waves

Wave are rather different : match rate is between 10 and 30 %

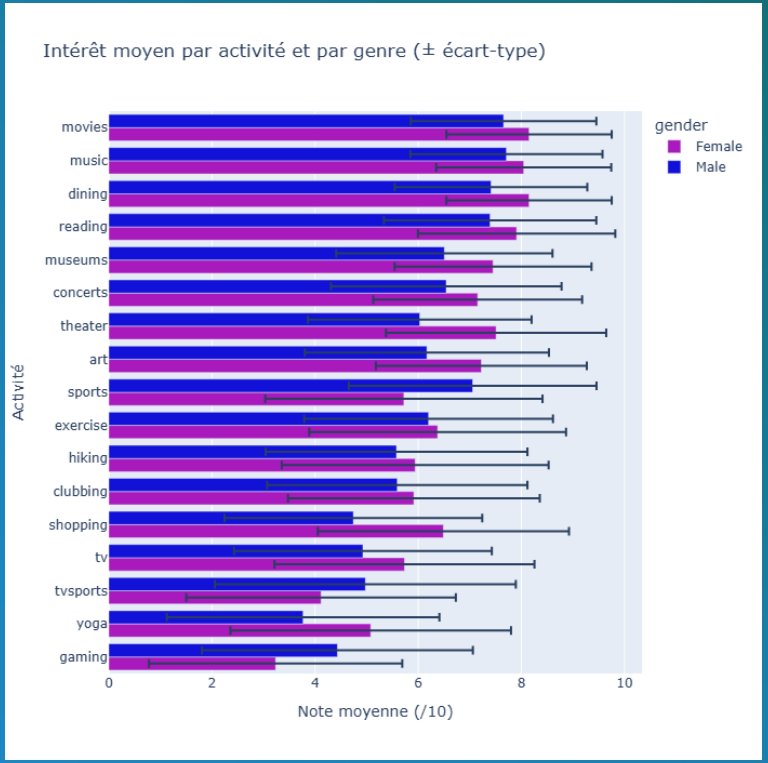
Men decision is pretty higher than women's



2 EDA of speed daters

Interests per activity

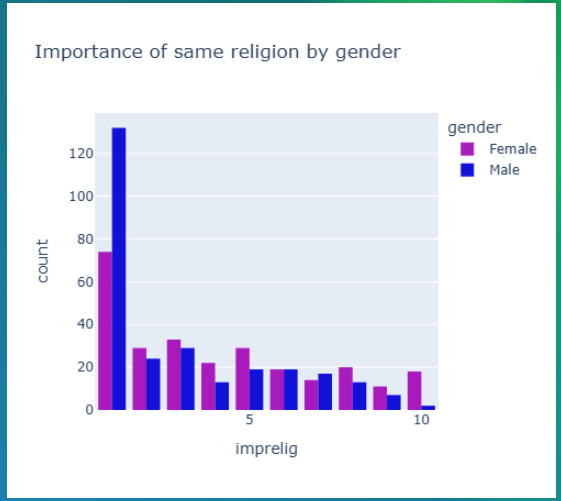
Well balanced between men and women

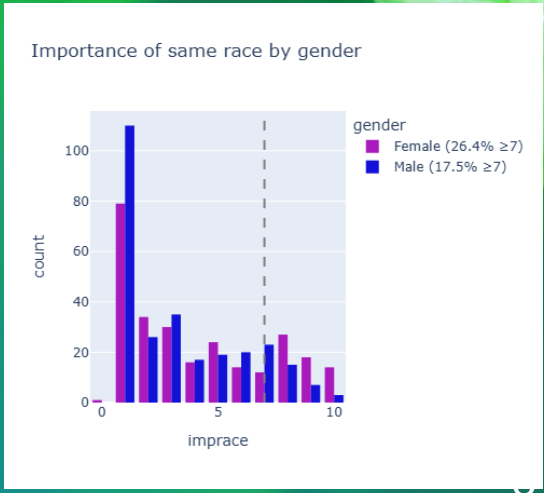


Importance race and religion :

Mainly no importance for men,

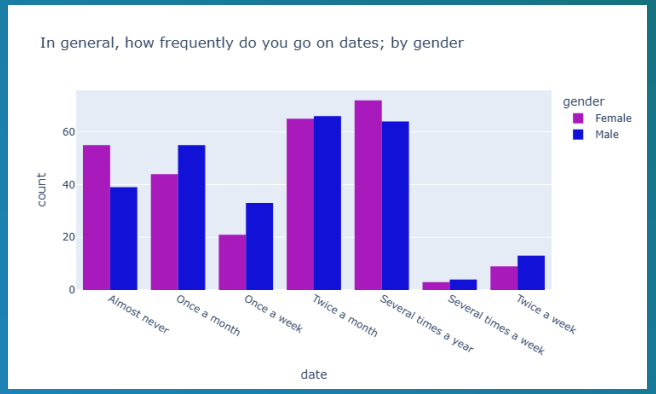
women : 25% have rated more than 7,

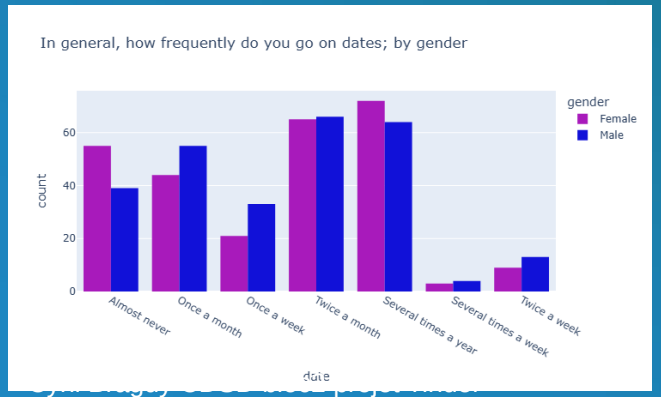




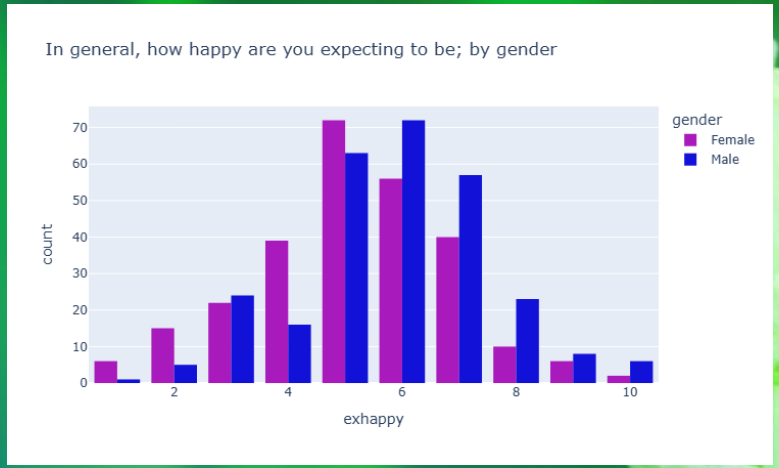
2 EDA of speed daters

Both go out/ on dates at same frequency

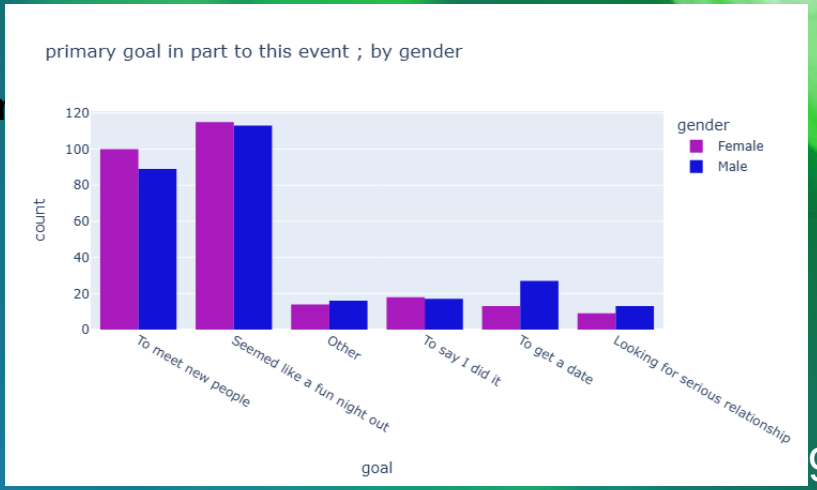




men are more expecting than women



But both are aligned on prim goal



3 Analysis

Main goal to answer :

What drives people to accept a second date.

One will use intermediate questions :

Q1 : What are the least desirable attributes in a male partner?
Does this differ for female partners?

Q2: How important do people think attractiveness is in potential mate selection vs. its real impact?

Q3: Are shared interests more important than a shared racial background?

Q4 : Can people accurately predict their own perceived value in the dating market?

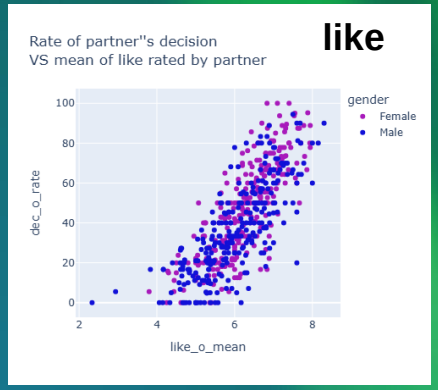
Q5: In terms of getting a second date, is it better to be someone's first speed date of the night or their last?

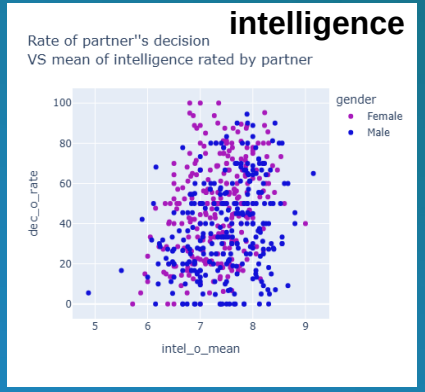
3 Analysis

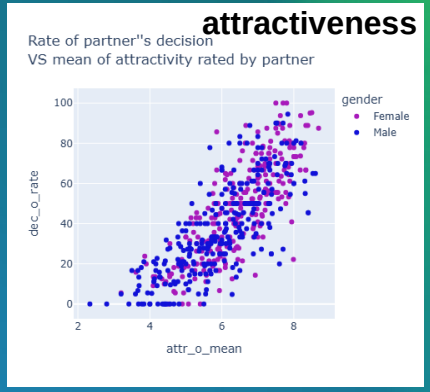
What are attributes impacting decision of a 2nd date ?
Does this differ for female partners? => dec_o_rate

According to scatter plots and correlations : Men who have partner's decision are rated by partner :

- **like**
- **attractiveness**
- **shared of interests**
- **fun**
  - **not intelligence**
  - **not ambition**
  - **not sincerity**







correlation between partner's decision and partner's ratings

	Male rated	Female rated
like_o	0,48	0,54
attractiveness	0,21	0,29
Shared interests	0,20	0,18
fun	0,19	0,16
Importance of race	-0,02	-0,01
Importance of religion	-0,03	-0,03
age	-0,03	-0,05
sincerity	-0,17	-0,25
ambition	-0,21	-0,18
intelligence	-0,24	-0,29

3 Analysis

How important do people think attractiveness is in potential mate selection vs. its real impact?

Synthesis :

Men, declare most looking for :

- attractiveness
- intelligence, ✗
- fun
- sincerity ✗

They decided most from :

- Like & attractiveness,
- shared interests,
- fun

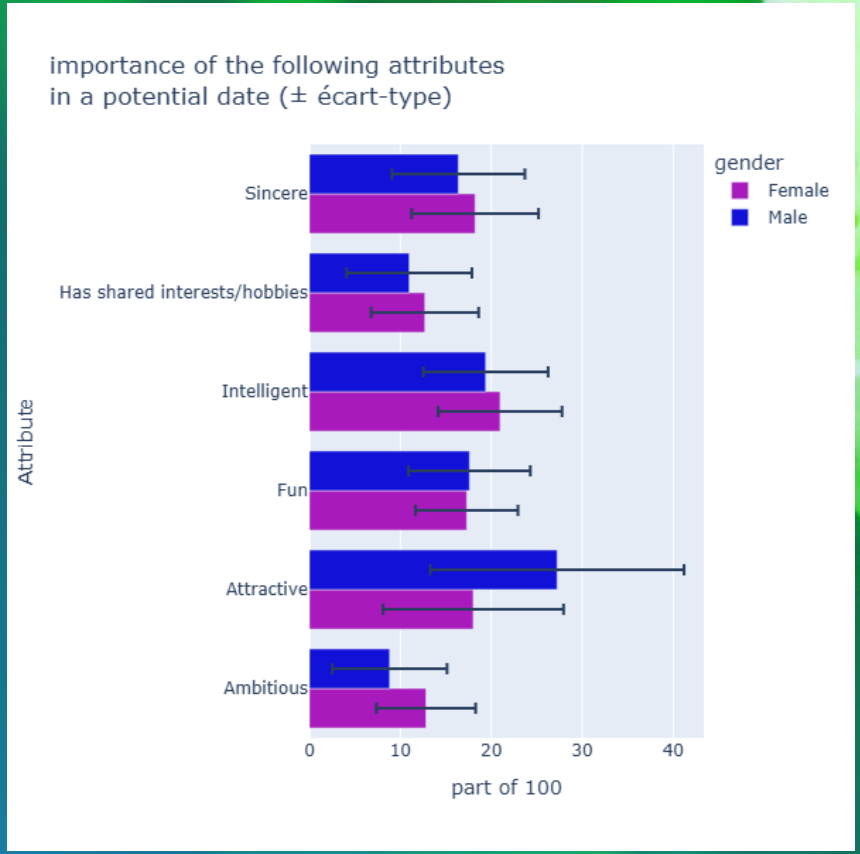
Women :

- Intelligence ✗
- sincerity, ✗
- fun
- attractiveness

- like, attractiveness
- shared interests,
- fun

Attributes declared by men are opposite to those correlated to their decision's trends : (from more to less important to make a decision Yes :)

whereas high perceived intelligence ambition and sincerity will cause a no-decision
same trend for women



3 Analysis

Are shared interests more important than a shared racial background?

Shared interests correlated at resp 20% and 18 %
to decision to have a next date

Shared racial background :

Compare importance on decision of :

- shar_o (importance of same race : num 1-10)
- samerace (the 2 people are of same race : bool)

with methods Chi2 and ROC_AUC (area under curve)

Samerace has a p-value very low, below 0.05 : the link with decision is close to uncorrelated

Both methods show that shared interests is more correlated than same_race to decision

	Male rated	Female rated
like	0,48	0,54
attractiveness	0,21	0,29
shared interests	0,20	0,18

Variable	φ / effect	AUC	Conclusion
shar (continous)	—	0.618	correlated
shar_bool (boolean)	0.175	0.589	a bit less correlated, some info lost
samerace	0.023	0.511	almost random link

3 Analysis

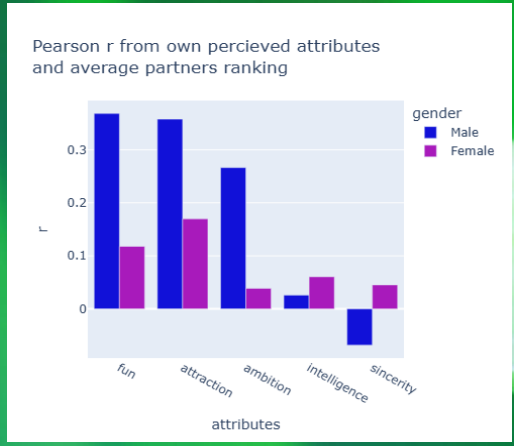
Can people accurately predict their own perceived value in the dating market?

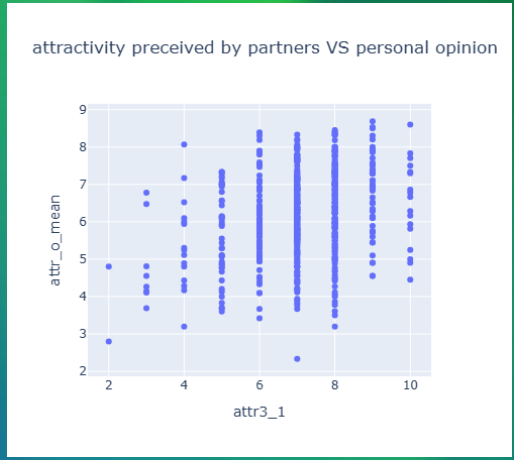
**One shall compare people's own vision (attr3_1 family)
and partner's ratings (attr_o family) with mean and std & correlation**

For Men, Moderate convergence (r about 0.3) is seen on :
- attractivity
- fun
Low convergence (0.15) is seen on
- ambition

for women, moderate/low convergence seen on attraction, and low on Fun , none on others attributes

But people are not at all able to auto evaluate on intelligence and even less on sincerity





3 Analysis

In terms of getting a second date, is it better to be someone's first speed date of the night or their last?

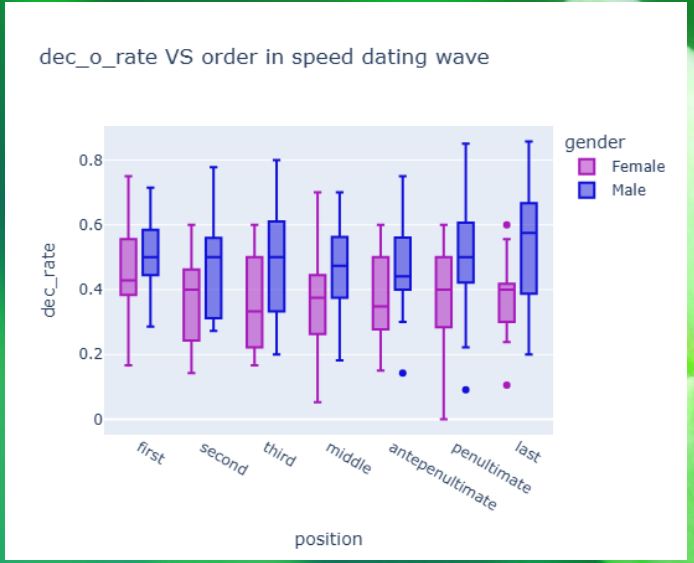
One shall look for correlation between date order of partner (order: integer) // nb of rounds and decision of him.er : dec_o (binary / boolean)

Graphical : no clear correlation is seen

compute biserial correlations between order_rate (numerical variable) and decision (boolean variable) :
correlation is very low, or null r “ 0,1 p-val>> 0,05

No clear link can be seen on the global data

Conclusion : no link with order in wave



	order_rate	
	r	P-value
dec	-0,014	0,18
dec_o	-0,013	0,2
match	-0,019	0,08

Global question

What makes people interested into each other to go on a second date together

The global conclusion is that :

Average match rate is around 16.5% ; this rate can change to up to 30% depending on wave

Main attributes impacting decision to request a second date are :
a high :
- like
- attraction
- shared interests,
- fun

a low : (for men)
- intelligence
- ambition
- sincerity

For Women :
- like
- attraction

in addition, almost no impact is seen on:

race, religion, age !! and field of study or career

People are looking for Fun when they date on Tinder !